

Mathematical Appendix for Structural Hawkes Ambiguity in Near-Critical Market Making

1 Purpose

This companion appendix records the proof details that would slow the main paper’s narrative. MESA means market-making excitation-structure ambiguity. The first sections cover the scalar amplification result, a Perron-visible multitype bridge, the finite-state Bellman layer used in the dynamic program, and a compact HJBI verification result for the truncated multitype game. The later sections spell out the extra conditions needed beyond truncation: robust subcriticality, weighted comparison, and local sensitivity of interior quote optimizers. The scope is deliberately narrow: the appendix covers only the model class used in the main paper and makes the assumptions needed for extensions explicit.

2 Notation Used Below

Symbol	Domain	Meaning
ρ	$[0, 1)$	Scalar branching ratio or Perron root $\rho(\Gamma)$, depending on context.
Γ	$\mathbb{R}_+^{d \times d}$	Multitype Hawkes branching matrix.
μ, λ	$\mathbb{R}_+^d$	Baseline intensity and current intensity vector.
r, ℓ	$\mathbb{R}_+^d$	Right and left Perron vectors, normalized by $\ell^\top r = 1$.
$E, E_{\text{rel}}, E_{\text{abs}}$	Matrix perturbations	Generic, relative-slack, and absolute branching-matrix perturbations.
$\varepsilon, \varepsilon_{\text{rel}}$	positive scalars	Absolute and relative ambiguity radii.
$V_{\text{risk}}, V_d, V_{\text{bi}}$	real-valued functions	Reduced risk proxies used to isolate the resolvent singularity.
q	$\{-Q, \dots, Q\}$	Inventory state in the Bellman and HJBI games.
K	compact subset of $\mathbb{R}_+^d$	Truncated continuous-intensity domain.
$U, \mathcal{A}$	compact or finite sets	Quote/no-quote controls and rectangular ambiguity set.
$L^{u, \Gamma}, \mathcal{L}^{u, \Gamma}$	generators	Finite-state and continuous-intensity controlled Hawkes generators.
W_v, κ	Lyapunov function and drift margin	Weighted intensity Lyapunov function and common subcriticality margin.
J_i, K_R	jump map and compact set	Untruncated Hawkes mark- i jump map and Lyapunov sublevel truncation set $\{W_v \leq R\}$.
U, V, F	weighted viscosity functions	Generic subsolution, supersolution, and solution candidate in the comparison theorem.
δ, θ	quote vector and structural parameter	Local quote action and finite-dimensional branching/ambiguity parameter in the optimizer sensitivity theorem.
$\mathfrak{H}, \delta^*$	reduced Hamiltonian and optimizer	Robust local Hamiltonian after Nature's minimization and its interior quote maximizer.
T, g, r	finite horizon, payoff, reward	Terminal time, terminal payoff, and running reward.

Unless a theorem says otherwise, asymptotic constants are taken uniformly over the compact regularity classes stated there. The same letters may denote a scalar object in the scalar section and its multitype analogue later; each section restates the needed domain.

3 Scalar Hawkes Setup

Let N_t be a stationary linear Hawkes process with exponential kernel

$$\lambda_t = \mu + \int_0^{t^-} \beta \rho e^{-\beta(t-s)} dN_s, \quad 0 \leq \rho < 1, \quad \mu > 0, \quad \beta > 0.$$

Here ρ is the branching ratio. The stationary mean intensity is

$$\bar{\lambda}(\rho) = \frac{\mu}{1 - \rho}.$$

We use a reduced market-making risk proxy

$$V_{\text{risk}}(\rho) = C(1 - \rho)^{-a}, \quad C > 0,$$

where $a = 2$ matches the normalized-throughput variance convention used for the square-law statement in the main paper. If stationary throughput is modeled separately in the risk term, a can be larger; the proofs keep that dependence explicit.

Assumption A.1 (Local quote sensitivity). *For inventory q in a compact grid and quote controls in a compact interval, the robust half-spread correction is locally linear in the worst-case increase of V_{risk} :*

$$\psi_\varepsilon(\rho) = K \left[\sup_{\tilde{\rho} \in \mathcal{U}_\varepsilon(\rho)} V_{\text{risk}}(\tilde{\rho}) - V_{\text{risk}}(\rho) \right] + o \left(\sup_{\tilde{\rho} \in \mathcal{U}_\varepsilon(\rho)} |V_{\text{risk}}(\tilde{\rho}) - V_{\text{risk}}(\rho)| \right),$$

with $K > 0$ independent of ρ near one.

This assumption is the scalar version of differentiating the first-order condition for an interior quote optimizer. The optimizer-sensitivity theorem below gives one sufficient route: a uniformly negative quote-space second derivative together with a smooth active worst-case selector for the robust Hamiltonian.

4 Resolvent and Variance Scaling

Lemma A.2 (Scalar resolvent). *For scalar Hawkes branching ratio $\rho < 1$,*

$$(1 - \rho)^{-1}$$

is the scalar Hawkes resolvent. Its square scales as $(1 - \rho)^{-2}$.

Proof. The stationary mean equation is $\bar{\lambda} = \mu + \rho \bar{\lambda}$. Therefore $\bar{\lambda} = (1 - \rho)^{-1} \mu$. The linear response of total descendants of one immigrant is the geometric series $\sum_{n \geq 0} \rho^n = (1 - \rho)^{-1}$, proving the resolvent identity. Squaring the leading response gives $(1 - \rho)^{-2}$. $\square$

Lemma A.3 (Derivative of the risk proxy). *Let $V_{\text{risk}}(\rho) = C(1 - \rho)^{-a}$ for $a > 0$. Then*

$$\frac{d}{d\rho} V_{\text{risk}}(\rho) = aC(1 - \rho)^{-(a+1)}.$$

Proof. Direct differentiation gives $d(1 - \rho)^{-a}/d\rho = a(1 - \rho)^{-(a+1)}$. $\square$

5 Ambiguity Normalizations

We distinguish two ambiguity sets. This is a small modeling choice with a large near-critical consequence, because the normalization changes the singular exponent.

Relative critical-slack ambiguity.

$$\mathcal{U}_\varepsilon^{\text{rel}}(\rho) = \{\tilde{\rho} : 0 \leq \tilde{\rho} \leq \rho + \varepsilon(1 - \rho), \tilde{\rho} < 1\}.$$

Absolute branching-ratio ambiguity.

$$\mathcal{U}_\varepsilon^{\text{abs}}(\rho) = \{\tilde{\rho} : 0 \leq \tilde{\rho} \leq \rho + \varepsilon, \tilde{\rho} < 1\}.$$

Theorem A.4 (Relative-slack amplification). *Fix $a > 0$ and $0 < \varepsilon < 1$. Under relative critical-slack ambiguity,*

$$\sup_{\tilde{\rho} \in \mathcal{U}_\varepsilon^{\text{rel}}(\rho)} V_{\text{risk}}(\tilde{\rho}) - V_{\text{risk}}(\rho) = C((1 - \varepsilon)^{-a} - 1)(1 - \rho)^{-a}.$$

If the local quote-sensitivity assumption holds uniformly near the reference regime, then, for small ε ,

$$\psi_\varepsilon(\rho) = KaC\varepsilon(1 - \rho)^{-a} + O(\varepsilon^2(1 - \rho)^{-a}) + o(\varepsilon(1 - \rho)^{-a}).$$

Within this reduced proxy, in the normalized-throughput convention $a = 2$, the premium has the square law $\psi_\varepsilon(\rho) = \Theta(\varepsilon(1 - \rho)^{-2})$.

Proof. The risk proxy is increasing in ρ , so Nature chooses $\tilde{\rho}^* = \rho + \varepsilon(1 - \rho)$. Then

$$1 - \tilde{\rho}^* = 1 - \rho - \varepsilon(1 - \rho) = (1 - \varepsilon)(1 - \rho).$$

Substitution gives

$$V_{\text{risk}}(\tilde{\rho}^*) - V_{\text{risk}}(\rho) = C(1 - \rho)^{-a}((1 - \varepsilon)^{-a} - 1).$$

The Taylor expansion $(1 - \varepsilon)^{-a} = 1 + a\varepsilon + O(\varepsilon^2)$ gives the small- ε expression. The quote-premium statement follows from local quote sensitivity. Since all constants are positive and independent of ρ near one, the result is $\Theta(\varepsilon(1 - \rho)^{-a})$. $\square$

Theorem A.5 (Absolute-ambiguity derivative law). *Fix $a > 0$ and suppose $\varepsilon = o(1 - \rho)$. Under absolute ambiguity,*

$$\sup_{\tilde{\rho} \in \mathcal{U}_\varepsilon^{\text{abs}}(\rho)} V_{\text{risk}}(\tilde{\rho}) - V_{\text{risk}}(\rho) = aC\varepsilon(1 - \rho)^{-(a+1)} + O(\varepsilon^2(1 - \rho)^{-(a+2)}).$$

If the local quote-sensitivity assumption holds uniformly near the reference regime, then, within this reduced proxy, the normalized-throughput convention $a = 2$ gives the premium order $\Theta(\varepsilon(1 - \rho)^{-3})$.

Proof. Again Nature chooses $\tilde{\rho}^* = \rho + \varepsilon$ when this remains below one. Taylor expanding around ρ ,

$$V_{\text{risk}}(\rho + \varepsilon) - V_{\text{risk}}(\rho) = \varepsilon V'_{\text{risk}}(\rho) + \frac{\varepsilon^2}{2} V''_{\text{risk}}(\xi)$$

for some $\xi \in (\rho, \rho + \varepsilon)$. By the derivative lemma, $V'_{\text{risk}}(\rho) = aC(1 - \rho)^{-(a+1)}$. Also $V''_{\text{risk}}(\xi) = a(a+1)C(1 - \xi)^{-(a+2)}$. The condition $\varepsilon = o(1 - \rho)$ implies $1 - \xi$ is comparable to $1 - \rho$, giving the stated remainder. Local quote sensitivity transfers the scaling to ψ_ε . $\square$

6 Multitype Perron Extension

Proposition A.6 (First-order Perron perturbation). *Let Γ be nonnegative and irreducible with simple Perron root $\rho < 1$, normalized left/right Perron vectors $\ell^\top r = 1$, spectral gap at least $g_0 > 0$, uniformly bounded Perron eigenprojection, and uniformly bounded reduced resolvent on the non-Perron spectral subspace. Then, for sufficiently small perturbations E ,*

$$\rho(\Gamma + E) - \rho(\Gamma) = \ell^\top E r + O(\|E\|_F^2/g_0).$$

Among unconstrained Frobenius-unit directions, the first-order maximizer is $E^ = \ell r^\top / \|\ell r^\top\|_F$. Feasibility of a scaled perturbation inside a particular ambiguity set must be checked separately.*

Proof. The first statement is the standard analytic perturbation expansion for a simple eigenvalue. The derivative of the Perron root in direction E is $\ell^\top E r = \langle E, \ell r^\top \rangle_F$. Under a Frobenius norm constraint, Cauchy–Schwarz gives

$$\langle E, \ell r^\top \rangle_F \leq \|E\|_F \|\ell r^\top\|_F,$$

with equality when E is aligned with $\ell r^\top$. $\square$

Remark A.7. *This proposition explains the Perron-aligned adversary used in the experiments when the leading eigenvalue is simple and well separated. It also explains why near-degenerate spectra are a real scope condition rather than a nuisance parameter: the lower bound can weaken when Perron regularity fails.*

Theorem A.8 (Finite-dimensional Perron-visible ambiguity premium). *Let $\Gamma \in \mathbb{R}_+^{d \times d}$ be irreducible, primitive, and subcritical with simple Perron root $\rho < 1$. Assume a uniform spectral gap*

$$\rho - \max_{j>1} |\lambda_j(\Gamma)| \geq g_0 > 0$$

and uniformly bounded Perron eigenprojection and reduced-resolvent conditioning. Let $r, \ell > 0$ be normalized by $\ell^\top r = 1$. For fixed $\mu, w \in \mathbb{R}_+^d$, define

$$V_d(\Gamma) = C \left(w^\top (I - \Gamma)^{-1} \mu \right)^2, \quad C > 0,$$

and assume Perron visibility

$$A(\Gamma) := (w^\top r)(\ell^\top \mu) \geq a_0 > 0.$$

Then

$$V_d(\Gamma) = C A(\Gamma)^2 (1 - \rho)^{-2} + O((1 - \rho)^{-1}).$$

If a feasible Perron-aligned relative-slack perturbation satisfies

$$\rho(\Gamma + E_{\text{rel}}) = \rho + \varepsilon_{\text{rel}}(1 - \rho) + O(\varepsilon_{\text{rel}}^2(1 - \rho)),$$

and the perturbation path remains in the same compact Perron-regular class with Perron visibility bounded away from zero and no first-order cancellation, then

$$V_d(\Gamma + E_{\text{rel}}) - V_d(\Gamma) = \Theta(\varepsilon_{\text{rel}}(1 - \rho)^{-2}).$$

If a feasible Perron-aligned absolute perturbation satisfies

$$\rho(\Gamma + E_{\text{abs}}) = \rho + \varepsilon + O(\varepsilon^2), \quad \varepsilon = o(1 - \rho),$$

and the same Perron-regularity and visibility conditions remain in force, then

$$V_d(\Gamma + E_{\text{abs}}) - V_d(\Gamma) = \Theta(\varepsilon(1 - \rho)^{-3}).$$

Under the local quote-sensitivity assumption, the robust half-spread premium has the same respective orders.

Proof. By the Riesz projection formula for the isolated Perron eigenvalue,

$$(I - \Gamma)^{-1} = \frac{r\ell^\top}{1 - \rho} + R, \quad \|R\| \leq C(g_0),$$

where the constant is uniform on the Perron-regular class. Therefore

$$w^\top (I - \Gamma)^{-1} \mu = \frac{(w^\top r)(\ell^\top \mu)}{1 - \rho} + O(1).$$

Squaring gives the displayed expansion for V_d because Perron visibility keeps the leading coefficient bounded away from zero.

For perturbations, the first-order Perron expansion gives

$$\rho(\Gamma + E) - \rho(\Gamma) = \ell^\top E r + O(\|E\|_F^2 / g_0).$$

The Perron eigenvectors and the visibility coefficient vary by lower-order terms under the stated local perturbations. Expanding $(1 - \rho - \Delta\rho)^{-2} - (1 - \rho)^{-2}$ gives the relative-slack order when $\Delta\rho = \varepsilon_{\text{rel}}(1 - \rho) + O(\varepsilon_{\text{rel}}^2(1 - \rho))$, and the absolute-ambiguity derivative law when $\Delta\rho = \varepsilon + O(\varepsilon^2)$ with $\varepsilon = o(1 - \rho)$. The quote-premium statement follows from local quote sensitivity. $\square$

7 Finite-State Robust Bellman Bridge

This section records the finite-state control layer used by the implemented experiment package. It gives the robust dynamic-programming equation for the finite inventory, finite-regime approximation used numerically. By itself, this finite-state theorem is not the continuous-intensity multitype Hawkes HJBI bridge; that compact bridge is stated in the next section.

Let $\mathcal{X}$ be a finite state space whose elements encode inventory, cash-normalized wealth state, and a finite Hawkes-regime or intensity bin. Let $\mathcal{U}(x)$ be a nonempty finite set of admissible quote/no-quote actions at state x , and let $\mathcal{A} = \{\Gamma_1, \dots, \Gamma_M\}$ be a finite rectangular ambiguity set of stable branching matrices. For each $(u, \Gamma) \in \mathcal{U}(x) \times \mathcal{A}$, let $L^{u, \Gamma}(x, y)$ be a conservative generator on $\mathcal{X}$:

$$L^{u, \Gamma}(x, y) \geq 0 \quad (y \neq x), \quad L^{u, \Gamma}(x, x) = - \sum_{y \neq x} L^{u, \Gamma}(x, y).$$

Assume bounded running rewards $r(x, u, \Gamma)$ and bounded terminal payoff $g(x)$.

Theorem A.9 (Finite-state robust Hawkes-regime Bellman verification). *For horizon $T < \infty$, the rectangular robust market-making game*

$$\sup_u \inf_{\Gamma} \mathbb{E} \left[\int_t^T r(X_s, u_s, \Gamma_s) ds + g(X_T) \mid X_t = x \right],$$

where u_s and Γ_s are predictable Markov selectors from $\mathcal{U}(X_s)$ and $\mathcal{A}$, has a value $V(t, x)$. The value is the unique bounded solution of the finite HJBI system

$$-\partial_t V(t, x) = \max_{u \in \mathcal{U}(x)} \min_{\Gamma \in \mathcal{A}} \left[r(x, u, \Gamma) + \sum_{y \in \mathcal{X}} L^{u, \Gamma}(x, y) V(t, y) \right], \quad V(T, x) = g(x).$$

Moreover, deterministic Markov selectors attaining the displayed finite maximum and minimum are optimal. The corresponding discrete-time statement holds for a stochastic one-step kernel; for the explicit Euler kernel this requires Δt small enough that $I + \Delta t L^{u, \Gamma}$ has nonnegative entries and row sums one.

Proof. Because $\mathcal{X}$, $\mathcal{U}(x)$, and $\mathcal{A}$ are finite and rewards are bounded, the Hamiltonian

$$H_x(v) = \max_{u \in \mathcal{U}(x)} \min_{\Gamma \in \mathcal{A}} \left[r(x, u, \Gamma) + \sum_y L^{u, \Gamma}(x, y) v(y) \right]$$

is globally Lipschitz in $v \in \mathbb{R}^{|\mathcal{X}|}$. Indeed, for any v, w , the difference between two generator pairings is bounded by $2\Lambda \|v - w\|_\infty$, where $\Lambda = \max_{x, u, \Gamma} \sum_{y \neq x} L^{u, \Gamma}(x, y) < \infty$; taking finite maxima and minima preserves this Lipschitz constant. Hence the terminal value problem $-\dot{V} = H(V)$, $V(T) = g$, has a unique bounded classical solution by the Picard–Lindelof theorem for finite-dimensional ODEs.

For a small step Δt , conditioning on the first jump of the controlled Markov chain gives

$$V(t, x) = \max_u \min_\Gamma \left[r(x, u, \Gamma) \Delta t + \sum_y P_{\Delta t}^{u, \Gamma}(x, y) V(t + \Delta t, y) \right] + o(\Delta t),$$

where $P_{\Delta t}^{u, \Gamma} = I + \Delta t L^{u, \Gamma} + o(\Delta t)$. Subtract $V(t + \Delta t, x)$, divide by Δt , and let $\Delta t \downarrow 0$ to recover the HJBI system. Conversely, applying Dynkin’s formula to the unique solution under arbitrary Markov selectors and using the pointwise max-min inequality shows that the solution dominates every admissible controller against worst-case Nature and is achieved by any selectors attaining the finite maximum and minimum. Such selectors exist because the action and ambiguity sets are finite. The discrete-time statement follows by the same one-step conditioning argument without taking the limit. $\square$

Remark A.10. *This theorem supports the finite-scenario Bellman layer used in the code. If the scenario set contains Perron-aligned relative-slack perturbations, the finite game can express the robust quote/no-quote response to those regimes. The theorem does not prove the continuum multitype Hawkes HJBI or differentiability of the untruncated optimizer with respect to Γ . The next section gives the compact continuous-intensity bridge.*

8 Compact Continuous-Intensity HJBI Bridge

This section packages the compact-domain verification step, conditional on the stated compact-PDMP dynamic-programming and comparison hypotheses. The argument uses standard PDMP, stochastic-control, and viscosity-solution machinery [3, 2, 1]. The next two sections remove the projection through a Lyapunov/tightness argument and a weighted comparison theorem, and the consolidated theorem below records the resulting untruncated statement.

Let the state be $X_t = (q_t, \lambda_t)$, where $q_t \in \{-Q, \dots, Q\}$ is inventory and $\lambda_t \in K \subset \mathbb{R}_+^d$ is the vector of Hawkes intensities. The compact set K is interpreted as the numerical truncation domain. Between jumps,

$$d\lambda_t = \beta(\mu - \lambda_t) dt.$$

At a mark- i jump, inventory changes by a bounded increment $\zeta_i(q, u)$ and intensity is updated by a truncated Hawkes jump map

$$\lambda_t = \Pi_K(\lambda_{t-} + \beta \Gamma e_i),$$

where Π_K is a fixed Lipschitz projection into K . The market maker chooses $u \in U(q, \lambda)$ and Nature chooses $\Gamma \in \mathcal{A}$, where $U(q, \lambda)$ and $\mathcal{A}$ are compact. Let $\Lambda_i(q, \lambda, u, \Gamma)$ be the controlled mark

intensities after quote placement and censoring by inventory limits. Define the generator on smooth test functions φ by

$$\begin{aligned}\mathcal{L}^{u,\Gamma}\varphi(q,\lambda) &= \beta(\mu - \lambda) \cdot \nabla_\lambda \varphi(q,\lambda) \\ &\quad + \sum_{i=1}^d \Lambda_i(q,\lambda,u,\Gamma) [\varphi(q + \zeta_i(q,u), \Pi_K(\lambda + \beta \Gamma e_i)) - \varphi(q,\lambda)].\end{aligned}$$

Assumption A.11 (Compact Hawkes control regularity). *The sets K , $\mathcal{A}$, and $U(q,\lambda)$ are compact, with $U(q,\lambda)$ having a closed graph. The ambiguity set satisfies $\sup_{\Gamma \in \mathcal{A}} \rho(\Gamma) \leq \rho_{\max} < 1$. The functions $r(q,\lambda,u,\Gamma)$, $g(q,\lambda)$, and $\Lambda_i(q,\lambda,u,\Gamma)$ are bounded and continuous, and are uniformly Lipschitz in (q,λ) on each inventory slice. The jump map above remains inside K , and the total jump rate is uniformly bounded by $\bar{\Lambda}$. Admissible controllers are nonanticipative predictable controls; Nature's choice of Γ is rectangular and predictable over time. The compact PDMP game satisfies the dynamic-programming principle for this admissible class, and bounded viscosity sub/supersolutions of the compact HJBI satisfy the standard comparison principle.*

Theorem A.12 (Conditional compact continuous-intensity Hawkes HJBI verification). *Under compact Hawkes control regularity, the robust finite-horizon rectangular game*

$$V(t,q,\lambda) = \sup_u \inf_{\Gamma} \mathbb{E} \left[\int_t^T r(q_s, \lambda_s, u_s, \Gamma_s) ds + g(q_T, \lambda_T) \mid q_t = q, \lambda_t = \lambda \right]$$

has a bounded uniformly continuous value function. It is the unique bounded viscosity solution, for the compact admissible class specified above, on $[0, T] \times \{-Q, \dots, Q\} \times K$ of

$$-\partial_t V = \sup_{u \in U(q,\lambda)} \inf_{\Gamma \in \mathcal{A}} \left\{ r(q,\lambda,u,\Gamma) + \mathcal{L}^{u,\Gamma} V(q,\lambda) \right\}, \quad V(T,q,\lambda) = g(q,\lambda).$$

If, in addition, the value is C^1 in (t,λ) on each inventory slice and the displayed supremum and infimum admit measurable selectors $u^*(t,q,\lambda)$ and $\Gamma^*(t,q,\lambda)$ satisfying Isaacs' pointwise saddle condition, then these selectors verify the game value. In the absence of exact selectors, measurable η -optimizers are η -optimal.

Proof. The assumptions define a controlled piecewise-deterministic Markov process on the compact state space $\{-Q, \dots, Q\} \times K$. Bounded rates rule out explosion, and compactness with continuity gives measurable selectors for the relaxed one-step Hamiltonians. The assumed compact PDMP dynamic-programming principle gives

$$V(t,x) = \sup_u \inf_{\Gamma} \mathbb{E} \left[\int_t^{t+h} r(X_s, u_s, \Gamma_s) ds + V(t+h, X_{t+h}) \mid X_t = x \right]$$

for $x = (q,\lambda)$.

Let ϕ be a smooth test function touching V from above at (t,x) . Apply the dynamic programming principle with nearly optimal controls over $[t, t+h]$, use the first-jump expansion for the PDMP, subtract $\phi(t,x)$, divide by h , and let $h \downarrow 0$. This gives the viscosity subsolution inequality. The same argument with a test function touching from below, freezing any admissible control and allowing Nature to use a nearly minimizing response, gives the supersolution inequality.

For comparison, the Hamiltonian is proper and uniformly Lipschitz in the value argument because total jump rates are bounded by $\bar{\Lambda}$, rewards are bounded, and coefficients are uniformly Lipschitz on the compact domain. The assumed compact comparison principle therefore implies that

any bounded upper-semicontinuous subsolution is dominated by any bounded lower-semicontinuous supersolution with ordered terminal data. Perron's method and the dynamic programming principle give existence, hence uniqueness of the bounded viscosity solution.

If V is classically differentiable in (t, λ) , applying Dynkin's formula to $V(s, X_s)$ under arbitrary selectors and using the pointwise max-min inequalities proves that V bounds the payoff from above and below. Selectors attaining the saddle Hamiltonian make those inequalities equalities, so they verify the value. Compactness gives measurable η -optimizers when exact selectors are unavailable, yielding η -optimal controls. $\square$

Remark A.13. *This theorem records the compact truncated bridge conditional on the standard compact-PDMP dynamic-programming and comparison hypotheses stated above. The next theorem removes the compact intensity projection for stability and truncation tails. The weighted comparison section then gives the untruncated viscosity identification for pairs with a fixed weighted boundary condition at infinity, and the optimizer-sensitivity section isolates no-quote/boundary active-set differentiability as a separate nonsmooth control issue.*

9 Untruncated Lyapunov Bridge

This section removes the artificial compact intensity projection at the level of nonexplosion, moment control, and truncation error. It is not a universal comparison theorem for every unbounded HJBI; it supplies the missing Lyapunov/tightness component under an explicit common subcriticality condition.

Let $\lambda_t \in \mathbb{R}_+^d$ evolve without projection:

$$d\lambda_t = \beta(\mu - \lambda_t) dt, \quad \lambda_t \mapsto \lambda_t + \beta \Gamma e_i \quad \text{at a mark-}i \text{ jump.}$$

Controls may thin or censor mark intensities, but not amplify them beyond the Hawkes state: assume

$$0 \leq \Lambda_i(q, \lambda, u, \Gamma) \leq \lambda_i$$

for all admissible (q, λ, u, Γ) . This covers passive quote refusal and inventory censoring, which can reduce fill intensities but should not create order-flow arrivals ex nihilo.

Assumption A.14 (Common Lyapunov subcriticality and tail integrability). *There exist $v \in \mathbb{R}_+^d$ with strictly positive entries and $\kappa \in (0, 1]$ such that*

$$\Gamma^\top v \leq (1 - \kappa)v$$

componentwise for every $\Gamma \in \mathcal{A}$. Rewards and terminal payoffs have at most linear Lyapunov growth:

$$|r(q, \lambda, u, \Gamma)| + |g(q, \lambda)| \leq C(1 + v^\top \lambda)$$

uniformly over the finite inventory grid and admissible controls. Fix some $p > 1$. Since $\mathcal{A}$ is compact, the Hawkes jumps have uniformly bounded W_v -increments, so the p th Lyapunov moment calculation below is finite uniformly over $\mathcal{A}$.

Theorem A.15 (Conditional untruncated Hawkes Lyapunov bridge). *Under common Lyapunov subcriticality, the unprojected controlled Hawkes PDMP is nonexplosive for every predictable admissible controller and Nature selector. For*

$$W_v(\lambda) = 1 + v^\top \lambda,$$

its generator satisfies the uniform drift bound

$$\mathcal{L}^{u,\Gamma} W_v(q, \lambda) \leq \beta v^\top \mu - \beta \kappa v^\top \lambda \leq C_0 - C_1 W_v(\lambda)$$

for constants $C_0, C_1 > 0$ depending only on (β, μ, v, κ) . Consequently, for every finite horizon T and the $p > 1$ in the assumption,

$$\sup_{u,\Gamma} \mathbb{E}_{q,\lambda} \left[\sup_{0 \leq s \leq T} W_v(\lambda_s)^p \right] \leq C_{T,p} W_v(\lambda)^p,$$

and the finite-horizon value is finite in the weighted class $|V(t, q, \lambda)| \leq C_T W_v(\lambda)$. Moreover, if $\tau_R = \inf\{t : W_v(\lambda_t) \geq R\}$, then

$$\sup_{u,\Gamma} \mathbb{P}_{q,\lambda}(\tau_R \leq T) \leq \frac{C_{T,p} W_v(\lambda)^p}{R^p}.$$

Thus compact truncations on $\{W_v \leq R\}$ approximate the untruncated game in probability. For linearly growing payoffs, the exit-tail contribution is uniformly controlled by

$$\sup_{u,\Gamma} \mathbb{E}_{q,\lambda} \left[\sup_{s \leq T} W_v(\lambda_s) \mathbf{1}_{\{\tau_R \leq T\}} \right] \leq \frac{C_{T,p} W_v(\lambda)^p}{R^{p-1}},$$

so the compact-truncation payoff error vanishes locally uniformly as $R \uparrow \infty$.

Proof. For the affine test function W_v , the deterministic drift contributes

$$\beta(\mu - \lambda) \cdot \nabla_\lambda W_v = \beta v^\top \mu - \beta v^\top \lambda.$$

At a mark- i jump, W_v increases by $\beta v^\top \Gamma e_i$. Therefore

$$\mathcal{L}^{u,\Gamma} W_v = \beta v^\top \mu - \beta v^\top \lambda + \sum_{i=1}^d \Lambda_i(q, \lambda, u, \Gamma) \beta v^\top \Gamma e_i.$$

Using $\Lambda_i \leq \lambda_i$ and the common Lyapunov inequality gives

$$\sum_i \Lambda_i v^\top \Gamma e_i \leq \sum_i \lambda_i (\Gamma^\top v)_i \leq (1 - \kappa) \sum_i \lambda_i v_i.$$

Substitution yields

$$\mathcal{L}^{u,\Gamma} W_v \leq \beta v^\top \mu - \beta \kappa v^\top \lambda.$$

Since $W_v = 1 + v^\top \lambda$, this is the stated Foster–Lyapunov drift after absorbing constants.

Dynkin’s formula applied to $W_v(\lambda_{t \wedge \tau_R})$ gives a uniform first-moment bound up to τ_R . Gronwall’s inequality gives $\mathbb{E} W_v(\lambda_{t \wedge \tau_R}) \leq C_T W_v(\lambda_0)$ with constants independent of R , u , and Γ . Letting $R \uparrow \infty$ proves nonexplosion and the first-moment estimate.

For the p th moment, apply the generator to W_v^p . Each mark- i jump adds the bounded amount $c_i(\Gamma) = \beta v^\top \Gamma e_i$ to W_v . Taylor’s formula gives, for $W_v \geq 1$,

$$(W_v + c_i)^p - W_v^p \leq p c_i W_v^{p-1} + C_p W_v^{p-2}.$$

The deterministic drift contributes $p W_v^{p-1} \beta(v^\top \mu - v^\top \lambda)$, while

$$\sum_i \lambda_i c_i(\Gamma) \leq \beta(1 - \kappa) v^\top \lambda.$$

Thus, uniformly over controls and Nature,

$$\mathcal{L}^{u,\Gamma} W_v^p \leq C_p W_v^{p-1} - p\beta\kappa W_v^p \leq C'_p - c_p W_v^p$$

after increasing constants. Dynkin's formula and Gronwall yield $\mathbb{E}W_v(\lambda_{t\wedge\tau_R})^p \leq C_{T,p}W_v(\lambda_0)^p$ uniformly in R , u , and Γ .

For the maximal estimate, write the stopped semimartingale decomposition

$$W_v(\lambda_{t\wedge\tau_R}) = W_v(\lambda_0) + A_{t\wedge\tau_R} + M_{t\wedge\tau_R}.$$

The drift variation satisfies $|A_{t\wedge\tau_R}| \leq C \int_0^{t\wedge\tau_R} W_v(\lambda_s) ds$. The predictable quadratic variation of M is also bounded by $C \int_0^{t\wedge\tau_R} W_v(\lambda_s) ds$, because each mark- i jump changes W_v by the fixed amount $\beta v^\top \Gamma e_i$ and the total marked jump rate is at most $\sum_i \lambda_i$. Burkholder–Davis–Gundy with exponent p and Gronwall then give

$$\mathbb{E} \sup_{s \leq T} W_v(\lambda_{s\wedge\tau_R})^p \leq C_{T,p} W_v(\lambda_0)^p,$$

uniformly in R . Letting $R \uparrow \infty$ proves the displayed maximal moment bound. Markov's inequality then gives the exit bound for τ_R .

The value-growth statement follows by integrating the assumed linear growth of running rewards and terminal payoff against the moment bound. On $\{W_v \leq R\}$, the compact HJBI theorem applies to the truncated game. The p th maximal moment and Holder's inequality give

$$\mathbb{E} \left[\sup_{s \leq T} W_v(\lambda_s) \mathbf{1}_{\{\tau_R \leq T\}} \right] \leq \left(\mathbb{E} \sup_{s \leq T} W_v(\lambda_s)^p \right)^{1/p} \mathbb{P}(\tau_R \leq T)^{1-1/p},$$

which is bounded by $C_{T,p} W_v(\lambda_0)^p / R^{p-1}$. This controls the payoff contribution from paths that leave the truncation set before T , proving convergence in probability and tail-controlled approximation for linearly growing payoffs. $\square$

Remark A.16. *If $\mathcal{A}$ is finite and each matrix is strictly subcritical, the common vector condition is easy to check by linear programming. For a general compact ambiguity set, it is a substantive robust-stability assumption, stronger than pointwise $\rho(\Gamma) < 1$. This is the tradeoff required by a uniform untruncated theorem.*

Remark A.17. *Combining the compact HJBI theorem with the Lyapunov bridge gives an untruncated value approximation theorem in the weighted class: solve on $\{W_v \leq R\}$, stop at the boundary, and let $R \uparrow \infty$. The next section adds the corresponding viscosity comparison theorem in the W_v -weighted class with an explicit boundary condition at infinity.*

10 Weighted Comparison and Global Untruncated HJBI

This section states a sufficient comparison condition for the unbounded-domain HJBI bridge, following the standard viscosity comparison strategy with a weighted barrier and a doubling-of-variables argument [1, 2]. It is deliberately explicit: without a weighted boundary condition at infinity, unbounded intensity states can admit spurious solutions.

Assumption A.18 (Weighted comparison regularity). *In addition to common Lyapunov subcriticality, assume:*

1. The ambiguity set $\mathcal{A}$ is compact, and the control set is either a fixed compact set or a compact correspondence $U(q, \lambda)$ continuous in Hausdorff distance on compact λ -sets; measurable η -selectors exist for the local Hamiltonian.

2. The controlled rates $\Lambda_i(q, \lambda, u, \Gamma)$ are continuous, bounded above by λ_i , and locally Lipschitz in λ , uniformly on compact subsets and uniformly over (u, Γ) .

3. The jump maps

$$J_i(q, \lambda, u, \Gamma) = (q + \zeta_i(q, u), \lambda + \beta \Gamma e_i)$$

are Lipschitz in λ and satisfy

$$W_v(J_i(q, \lambda, u, \Gamma)) \leq C_J W_v(\lambda).$$

4. Rewards and terminal payoffs are continuous and have at most linear W_v -growth.

5. Subsolutions U and supersolutions V are compared in the weighted class

$$|U(t, q, \lambda)| + |V(t, q, \lambda)| \leq C W_v(\lambda)$$

and satisfy the boundary condition at infinity

$$\limsup_{R \rightarrow \infty} \sup_{\substack{t \in [0, T], q \\ W_v(\lambda) \geq R}} \frac{U(t, q, \lambda) - V(t, q, \lambda)}{W_v(\lambda)} \leq 0.$$

Theorem A.19 (Sufficient weighted-comparison bridge for the untruncated Hawkes HJBI). *Under weighted comparison regularity, let U be an upper-semicontinuous viscosity subsolution and V a lower-semicontinuous viscosity supersolution of the untruncated robust Hawkes HJBI*

$$-\partial_t F = \sup_{u \in U(q, \lambda)} \inf_{\Gamma \in \mathcal{A}} \left\{ r(q, \lambda, u, \Gamma) + \mathcal{L}^{u, \Gamma} F(q, \lambda) \right\}$$

on $[0, T] \times \{-Q, \dots, Q\} \times \mathbb{R}_+^d$. If $U(T, q, \lambda) \leq V(T, q, \lambda)$ for all (q, λ) and the weighted boundary condition above holds, then

$$U(t, q, \lambda) \leq V(t, q, \lambda)$$

for every (t, q, λ) . Consequently, any two candidates in the same weighted-boundary subclass are ordered by comparison. Any compact-truncation limit satisfying this boundary condition is therefore uniquely identified, and the truncated solutions converge locally uniformly to it as $R \uparrow \infty$.

Proof. Fix $\delta > 0$ and define the strict supersolution

$$V^\delta(t, q, \lambda) = V(t, q, \lambda) + \delta e^{a(T-t)} W_v(\lambda),$$

with a chosen large enough that the Lyapunov drift dominates the linear growth constants in the Hamiltonian. The Foster–Lyapunov inequality gives

$$-\partial_t \{\delta e^{a(T-t)} W_v\} - \sup_u \inf_\Gamma \mathcal{L}^{u, \Gamma} \{\delta e^{a(T-t)} W_v\} \geq c \delta W_v(\lambda) - C \delta$$

for some $c > 0$ after increasing a . Thus V^δ is a strict supersolution outside a fixed compact set, and the weighted boundary condition implies that any positive maximum of $U - V^\delta$ occurs on $[0, T] \times \{-Q, \dots, Q\} \times K_R$ for some compact Lyapunov sublevel set $K_R = \{W_v \leq R\}$.

On this compact set, apply the standard doubling-of-variables argument to

$$U(t, q, \lambda) - V^\delta(s, q, \lambda') - \frac{|t - s|^2}{2\alpha} - \frac{|\lambda - \lambda'|^2}{2\varepsilon}.$$

The finite inventory coordinate is discrete, so at a positive maximum it can be matched by taking the same q ; otherwise the maximum over finitely many inventory states is handled componentwise. The viscosity inequalities at the maximizers, together with compactness of $U(q, \lambda)$ and $\mathcal{A}$, give nearly maximizing/minimizing controls and ambiguity selectors. Continuity and local Lipschitz regularity of rates, rewards, and jump maps control the drift and jump terms. The nonlocal terms are handled by the viscosity nonlocal doubling inequality:

$$U(t, J_i(q, \lambda, u, \Gamma)) - V^\delta(s, J_i(q, \lambda', u, \Gamma))$$

is bounded by the same doubled maximum plus an $O(|\lambda - \lambda'|^2/\varepsilon) + o(1)$ error, because each J_i is Lipschitz on K_R . Letting $\alpha, \varepsilon \downarrow 0$ yields a contradiction to strict positivity of the maximum unless $\sup(U - V^\delta) \leq 0$.

Therefore $U \leq V^\delta$ for every $\delta > 0$. Letting $\delta \downarrow 0$ proves comparison. Existence follows by solving the compactly truncated HJBI on K_R , extending with stopped dynamics, and using the Lyapunov exit estimate from the previous theorem to pass to the limit. Comparison identifies every subsequential limit satisfying the same weighted boundary condition at infinity, so the convergence is locally uniform and the limit is the unique viscosity solution in that fixed-boundary subclass. $\square$

Remark A.20. *This theorem records the untruncated HJBI comparison bridge for the model class in this appendix: common Lyapunov subcriticality with $p > 1$ tail control, linear-growth rewards, compact controls/ambiguity, and a fixed weighted boundary behavior at infinity. It still does not prove differentiability of boundary or no-quote active sets, nor does it cover state-dependent, power-law, or empirically learned Hawkes kernels without rechecking the same weighted comparison assumptions.*

11 Interior Quote Optimizer Sensitivity

The HJBI theorems above identify the value for the stated control class. This section records the additional local condition needed to connect the robust Hamiltonian to differentiable quote rules as the branching matrix or ambiguity parameter changes. It covers the smooth interior region of the implemented quote map; no-quote boundaries and active-set switches are intentionally handled as nonsmooth cases.

Fix a state/adjoint tuple z collecting (t, q, λ) and the value increments entering the local HJBI Hamiltonian. Let $\theta \in \Theta \subset \mathbb{R}^m$ be a finite-dimensional parameterization of either Γ itself or an ambiguity set centered at Γ . For a quote vector $\delta \in D \subset \mathbb{R}^k$, define the reduced robust Hamiltonian

$$\mathfrak{H}(\delta, \theta; z) = \inf_{\Gamma' \in \mathcal{A}(\theta)} \mathcal{H}(\delta, \Gamma'; z),$$

where $\mathcal{H}$ is the local reward-plus-generator expression in the HJBI.

Assumption A.21 (Smooth interior robust optimizer). *For the fixed z , assume there are neighborhoods of (δ_0, θ_0) such that:*

1. D is open around δ_0 , and the admissible quote constraints are inactive in that neighborhood.
2. Nature has a unique active minimizer $\Gamma^\dagger(\delta, \theta; z)$, and this selector is C^1 in (δ, θ) .

3. The reduced Hamiltonian $\mathfrak{H}$ is C^2 in δ and has continuous mixed derivative $D_{\delta\theta}^2 \mathfrak{H}$.

4. The quote objective is uniformly strictly concave:

$$D_{\delta\delta}^2 \mathfrak{H}(\delta, \theta; z) \preceq -mI_k$$

for some $m > 0$ throughout the neighborhood.

5. δ_0 is the unique local maximizer at θ_0 , so $D_\delta \mathfrak{H}(\delta_0, \theta_0; z) = 0$.

Theorem A.22 (Interior optimizer differentiability). *Under the smooth interior robust optimizer assumption, there exists a neighborhood N of θ_0 and a unique interior maximizer $\delta^*(\theta; z)$ in the corresponding quote neighborhood. The map $\theta \mapsto \delta^*(\theta; z)$ is C^1 on N , and*

$$D_\theta \delta^*(\theta_0; z) = - \left[D_{\delta\delta}^2 \mathfrak{H}(\delta_0, \theta_0; z) \right]^{-1} D_{\delta\theta}^2 \mathfrak{H}(\delta_0, \theta_0; z).$$

In the one-dimensional half-spread case, if e is a Perron-adverse structural direction and

$$D_{\delta\theta}^2 \mathfrak{H}(\delta_0, \theta_0; z)e \geq 0,$$

then $D_\theta \delta^*(\theta_0; z)e \geq 0$; the robust interior half-spread widens to first order in that adverse direction.

Proof. By the smooth active-minimizer assumption and the envelope/Danskin theorem, the reduced Hamiltonian has the stated derivatives in a neighborhood of (δ_0, θ_0) . Define

$$G(\delta, \theta) = D_\delta \mathfrak{H}(\delta, \theta; z).$$

The first-order condition at the reference point is $G(\delta_0, \theta_0) = 0$. The derivative of G with respect to δ is $D_{\delta\delta}^2 \mathfrak{H}(\delta_0, \theta_0; z)$, which is nonsingular because it is negative definite. The implicit function theorem therefore gives a unique C^1 function $\delta^*(\theta; z)$ solving $G(\delta^*(\theta; z), \theta) = 0$ locally. Strict concavity makes that critical point the unique local maximizer. Differentiating the identity $G(\delta^*(\theta; z), \theta) = 0$ gives the displayed formula. In dimension one, the denominator $D_{\delta\delta}^2 \mathfrak{H}$ is negative, so the sign of the directional derivative agrees with the sign of $D_{\delta\theta}^2 \mathfrak{H}e$. This proves the widening statement. $\square$

Remark A.23. *The theorem is local by design. If the optimal quote hits a no-quote boundary, if a cap such as $\delta \leq \delta_{\max}$ binds, or if two worst-case Hawkes matrices exchange activity, the optimizer may have only one-sided directional derivatives or a kink. Those cases are consistent with the implemented liquidity-guard policy, but they should be reported as nonsmooth active-set behavior rather than as classical differentiability.*

12 Consolidated Multitype Control Theorem

The preceding sections are intentionally modular because compact verification, untruncated stability, comparison, and quote sensitivity require different assumptions. The theorem below gives a single citation point for the sufficient conditions assembled above.

Theorem A.24 (Consolidated conditional multitype Hawkes control bridge). *Consider the finite-horizon robust market-making game with finite inventory grid, exponential d -type Hawkes intensity state $\lambda \in \mathbb{R}_+^d$, compact quote/no-quote control graph, compact rectangular ambiguity set $\mathcal{A}$, controlled mark rates bounded by the Hawkes intensity state, and continuous rewards and terminal payoff with at most linear W_v -growth. Suppose common Lyapunov subcriticality with the $p > 1$ tail-integrability condition and weighted comparison regularity hold, and compare candidate viscosity solutions only in a W_v -weighted subclass with the same stated boundary condition at infinity. Then:*

1. the controlled Hawkes PDMP is nonexplosive for every admissible controller and Nature selector;
2. the robust finite-horizon value is finite in the W_v -weighted class;
3. the value is the unique viscosity solution, in that weighted class and with the fixed stated boundary condition at infinity, of the untruncated multitype Hawkes HJBI equation;
4. compact Lyapunov-sublevel truncations converge locally uniformly to the untruncated value as the truncation radius tends to infinity;
5. if the untruncated value is classically differentiable, has the same W_v -growth, its generator is integrable under the selected localized dynamics, and the Isaacs saddle selectors exist with admissible localization limits, then those selectors verify the untruncated game value; and
6. on smooth interior quote regions satisfying the smooth interior robust optimizer assumption, with inactive quote constraints, a unique C^1 active worst-case structural selector, and a C^2 reduced Hamiltonian with continuous mixed derivative, uniformly negative quote Hessian, and a unique local maximizer—the robust quote map is C^1 in the branching/ambiguity parameter and its derivative is given by the implicit-function formula above.

Proof. Items 1 and 2 follow from the Lyapunov stability theorem: the common subcritical vector gives the Foster–Lyapunov drift bound, nonexplosion, $p > 1$ finite weighted maximal moments, and finite linearly growing payoffs. Item 3 follows from the weighted comparison theorem, which proves comparison for upper-semicontinuous subsolutions and lower-semicontinuous supersolutions in the W_v -weighted class with the boundary condition at infinity; existence is obtained by the compact truncation construction. The boundary condition at infinity is a condition on the compared candidate solutions, not a consequence of pointwise subcriticality alone. Item 4 is the local-uniform convergence conclusion of the same comparison-plus-truncation argument, using stopped Lyapunov-sublevel games and the Lyapunov exit estimate to control paths that leave compact sublevel sets; it does not identify projected-boundary controls with stopped-boundary controls without this localization error bound. Item 5 is the classical verification argument localized at τ_R , followed by Dynkin’s formula, the max–min inequalities, the weighted moment bounds, and then $R \uparrow \infty$ under the stated generator integrability and selector localization assumptions. Item 6 is the interior optimizer differentiability theorem, with its smooth active-minimizer and strict concavity hypotheses included explicitly in the theorem statement. $\square$

13 Bivariate Perron Bridge

This section records the bivariate bridge theorem used by the manuscript. It is still a reduced-form risk-premium theorem, not a complete HJBI proof, but it upgrades the scalar statement to the smallest genuinely multitype Hawkes setting.

Let

$$\Gamma = \begin{pmatrix} \gamma_{11} & \gamma_{12} \\ \gamma_{21} & \gamma_{22} \end{pmatrix}$$

be nonnegative, irreducible, and subcritical with Perron root $\rho = \rho(\Gamma) < 1$. Let $r, \ell > 0$ be right/left Perron vectors normalized by $\ell^\top r = 1$. Let the second eigenvalue be η , and assume a fixed gap condition along a near-critical sequence,

$$\rho \uparrow 1, \quad |\rho - \eta| \geq g_0 > 0.$$

For two-type order flow with baseline $\mu \in \mathbb{R}_+^2$, the stationary intensity is

$$\bar{\lambda} = (I - \Gamma)^{-1}\mu.$$

For a reduced market-making risk direction $w \in \mathbb{R}_+^2$, define

$$V_{\text{bi}}(\Gamma) = C \left(w^\top (I - \Gamma)^{-1} \mu \right)^2, \quad C > 0.$$

We work on a compact Perron-regular class with uniformly bounded eigenprojection conditioning and Perron visibility bounded below:

$$A := (w^\top r)(\ell^\top \mu) \geq a_0 > 0.$$

Lemma A.25 (Bivariate resolvent decomposition). *Under the bivariate assumptions above,*

$$(I - \Gamma)^{-1} = \frac{r\ell^\top}{1 - \rho} + R_\eta, \quad \|R_\eta\| \leq C_\eta,$$

where C_η depends only on the spectral gap, vector normalization, eigenprojection conditioning, and a compact upper bound on $\|\Gamma\|$.

Proof. Because Γ is a real 2×2 matrix with simple Perron root and separated second eigenvalue, the spectral projection onto the Perron eigenspace is $P = r\ell^\top$. The complementary projection is $I - P$. On the Perron subspace, $(I - \Gamma)^{-1}$ has eigenvalue $(1 - \rho)^{-1}$. On the complementary one-dimensional subspace, the resolvent has eigenvalue $(1 - \eta)^{-1}$ times a bounded projection. Since $|\rho - \eta| \geq g_0$ and $\rho < 1$, the complementary term is uniformly bounded on the stated compact class. This gives the decomposition. $\square$

Lemma A.26 (Bivariate risk singularity). *Under the same assumptions,*

$$V_{\text{bi}}(\Gamma) = CA^2(1 - \rho)^{-2} + O((1 - \rho)^{-1}).$$

Consequently $V_{\text{bi}}(\Gamma) = \Theta((1 - \rho)^{-2})$ as $\rho \uparrow 1$.

Proof. By the resolvent decomposition,

$$w^\top (I - \Gamma)^{-1} \mu = \frac{(w^\top r)(\ell^\top \mu)}{1 - \rho} + w^\top R_\eta \mu = \frac{A}{1 - \rho} + O(1).$$

Squaring yields

$$\left(\frac{A}{1 - \rho} + O(1) \right)^2 = A^2(1 - \rho)^{-2} + O((1 - \rho)^{-1}) + O(1).$$

Since $A > 0$, the leading term is nonzero and proves the stated singularity. $\square$

Assumption A.27 (Perron-coefficient regularity). *For the local perturbation paths below, the Perron visibility coefficient*

$$A(\Gamma) := (w^\top r(\Gamma))(\ell(\Gamma)^\top \mu)$$

does not cancel the gap singularity. Concretely, for relative-slack paths we assume $A(\Gamma + E_{\text{rel}}) = A(\Gamma) + o(\varepsilon_{\text{rel}})$, and for absolute paths we assume $A(\Gamma + E_{\text{abs}}) = A(\Gamma) + o(\varepsilon)$. A sufficient special case is a spectral path that changes the Perron root while holding the Perron eigendirections fixed to first order.

Theorem A.28 (Perron-aligned bivariate relative-slack premium). *Let E_{rel} be a perturbation for which*

$$\rho(\Gamma + E_{\text{rel}}) = \rho + \varepsilon_{\text{rel}}(1 - \rho) + O(\varepsilon_{\text{rel}}^2(1 - \rho)),$$

and suppose $\Gamma + E_{\text{rel}}$ remains in the same compact bivariate Perron-regular class with Perron visibility bounded below and satisfying Perron-coefficient regularity. Then

$$V_{\text{bi}}(\Gamma + E_{\text{rel}}) - V_{\text{bi}}(\Gamma) = 2CA^2\varepsilon_{\text{rel}}(1 - \rho)^{-2} + O(\varepsilon_{\text{rel}}^2(1 - \rho)^{-2}) + o(\varepsilon_{\text{rel}}(1 - \rho)^{-2}) + O(\varepsilon_{\text{rel}}(1 - \rho)^{-1}).$$

In particular, the leading singular order is $\Theta(\varepsilon_{\text{rel}}(1 - \rho)^{-2})$. Under local quote sensitivity, if the finite ambiguity set includes this Perron-aligned scenario and that scenario is selected as worst case, then the half-spread has the same square-law order.

Proof. The bivariate risk singularity gives

$$V_{\text{bi}}(\Gamma) = CA^2(1 - \rho)^{-2} + O((1 - \rho)^{-1}).$$

The perturbed gap satisfies

$$1 - \rho(\Gamma + E_{\text{rel}}) = (1 - \varepsilon_{\text{rel}})(1 - \rho) + O(\varepsilon_{\text{rel}}^2(1 - \rho)).$$

Uniform Perron visibility and compactness keep the leading coefficient bounded above and below. Therefore the leading change is the same calculation as in the scalar relative-slack theorem with $a = 2$:

$$CA^2(1 - \rho)^{-2}\{(1 - \varepsilon_{\text{rel}})^{-2} - 1\} = 2CA^2\varepsilon_{\text{rel}}(1 - \rho)^{-2} + O(\varepsilon_{\text{rel}}^2(1 - \rho)^{-2}).$$

The lower bound follows because A is bounded away from zero and the Perron-coefficient regularity prevents first-order cancellation; the complementary resolvent term is lower order. Local quote sensitivity transfers the order to the half-spread correction. $\square$

Theorem A.29 (Perron-aligned bivariate absolute-ambiguity derivative law). *Let E_{abs} satisfy*

$$\rho(\Gamma + E_{\text{abs}}) = \rho + \varepsilon + O(\varepsilon^2), \quad \varepsilon = o(1 - \rho),$$

and suppose the perturbed matrices remain in the same bivariate Perron-regular class with Perron visibility bounded below and satisfying Perron-coefficient regularity. Then

$$V_{\text{bi}}(\Gamma + E_{\text{abs}}) - V_{\text{bi}}(\Gamma) = 2CA^2\varepsilon(1 - \rho)^{-3} + O(\varepsilon^2(1 - \rho)^{-4}) + o(\varepsilon(1 - \rho)^{-3}) + O(\varepsilon(1 - \rho)^{-2}).$$

In particular, the leading singular order is $\Theta(\varepsilon(1 - \rho)^{-3})$. Under local quote sensitivity, if the finite ambiguity set includes this Perron-aligned scenario and that scenario is selected as worst case, then the half-spread has the same derivative-law order.

Proof. Using the same leading expansion,

$$V_{\text{bi}}(\Gamma + E_{\text{abs}}) - V_{\text{bi}}(\Gamma) = CA^2 \left((1 - \rho - \varepsilon + O(\varepsilon^2))^{-2} - (1 - \rho)^{-2} \right) + O(\varepsilon(1 - \rho)^{-2}).$$

Since $\varepsilon = o(1 - \rho)$, a Taylor expansion of x^{-2} at $x = 1 - \rho$ gives

$$(1 - \rho - \varepsilon)^{-2} - (1 - \rho)^{-2} = 2\varepsilon(1 - \rho)^{-3} + O(\varepsilon^2(1 - \rho)^{-4}).$$

The $O(\varepsilon(1 - \rho)^{-2})$ contribution from the bounded complementary resolvent and the $o(\varepsilon(1 - \rho)^{-3})$ contribution from the Perron coefficient are lower order. Positivity of the aligned Perron coefficient gives the matching lower bound. Local quote sensitivity again transfers the order. $\square$

Remark A.30 (How to realize the perturbations). *The first-order Perron perturbation proposition gives a concrete construction: take E proportional to the Frobenius-normalized Perron direction $\ell r^\top$. Then*

$$\rho(\Gamma + tE) - \rho(\Gamma) = t\langle E, \ell r^\top \rangle_F + O(t^2),$$

so choosing $t = O(\varepsilon_{\text{rel}}(1 - \rho))$ gives the relative theorem and choosing $t = O(\varepsilon)$ gives the absolute theorem. This matches the Perron-aligned direction used in the matrix-resolvent experiments, up to the feasibility and normalization choices imposed there.

14 Conclusion

The square-law statement should be read with its normalization in view. When the ambiguity radius is measured relative to the remaining critical slack, or when the risk quantity has already been normalized, the reduced premium has the square-law order. When ε is an absolute branching-matrix radius, the robust premium inherits one additional derivative of the critical singularity. The same distinction holds in the bivariate Perron-visible setting. In that sense, this appendix puts the MESA claim into precise, testable scalar and bivariate theorem statements.

References

- [1] M. G. Crandall, H. Ishii, and P.-L. Lions. User’s guide to viscosity solutions of second order partial differential equations. *Bulletin of the American Mathematical Society*, 27(1):1–67, 1992. doi:10.1090/S0273-0979-1992-00266-5.
- [2] W. H. Fleming and H. M. Soner. *Controlled Markov Processes and Viscosity Solutions*. 2nd ed., Springer, 2006. doi:10.1007/0-387-31071-1.
- [3] M. H. A. Davis. *Markov Models and Optimization*. Chapman & Hall, 1993. ISBN 9780412314100.